

Cosets and a Standard Array

$$\mathbb{F}_q^n = \coprod (a + \mathcal{C}), \text{ one syndrome per row}$$

the code $\mathcal{C}$

$s = 000$	0	w_{00}	w_{01}	w_{02}	w_{03}	w_{04}	w_{05}	w_{06}	w_{07}
$s = 100$	e_1	w_{10}	w_{11}	w_{12}	w_{13}	w_{14}	w_{15}	w_{16}	w_{17}
$s = 010$	e_2	w_{20}	w_{21}	w_{22}	w_{23}	w_{24}	w_{25}	w_{26}	w_{27}
$s = 110$	e_3	w_{30}	w_{31}	w_{32}	w_{33}	w_{34}	w_{35}	w_{36}	w_{37}

Nearest-neighbor decoding

compute syndrome

choose row leader

subtract it

Rows are cosets; columns are translates of the same linear code.